

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 4: Quadratic Equations (Core-Basics Edition)

CHAPTER 4: QUADRATIC EQUATIONS

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) POWER / SQUARE

$x^2 = x \text{ times } x$. x^2 ko "x square" bolte hain.

2) EQUATION VS EXPRESSION

Expression: $2x^2 + 3x - 5$ (sirf likha).

Equation: $2x^2 + 3x - 5 = 0$ ('=' sign hai).

3) DEGREE

Degree = sabse bada power. x^2 wala = degree 2.

4) COEFFICIENT (a, b, c)

$ax^2 + bx + c$ mein $a = x^2$ ka, $b = x$ ka, $c = \text{constant}$.

$3x^2 - 5x + 2 \rightarrow a=3, b=-5, c=2$.

5) ROOT / ZERO / SOLUTION

Root = woh x jiske liye equation 0 ho jaye.

(Quadratic ke max 2 roots hote hain.)

6) FACTORISATION (splitting middle term)

ax^2+bx+c ko $(..)(..)$ mein todo.

Middle term b ko p,q mein todo: $p+q=b$, $p \times q = a \times c$.

7) SQUARE ROOT (sqrt)

$\sqrt{25}=5$ kyunki $5 \times 5=25$. Power ka ulta.

$\sqrt{D}$ ka matlab D ka square root.

8) ZERO PRODUCT RULE

Agar $A \times B = 0$, toh ya $A=0$ ya $B=0$ (ya dono).

Yahi factorisation method ka base hai.

9) PLUS-MINUS (+/-)

Quadratic formula mein +/- ka matlab DO answers:

ek + wala, ek - wala.

NOW: ASLI CHAPTER 4 SHURU

TOPIC 1: QUADRATIC EQUATION KYA HAI?

Standard form: $ax^2 + bx + c = 0$ ($a \neq 0$).
Degree 2 ki equation. Max 2 roots.

Quadratic: $x^2 - 5x + 6 = 0$, $2x^2 + 7x = 0$.
Not: $2x + 5 = 0$ (no x^2), $x^3 + 1 = 0$ (degree 3).

TOPIC 2: SOLVE BY FACTORISATION

EXAM ALERT: Factorisation se solve karna pakka aata hai.

Steps: middle term todo ($p+q=b$, $pq=ac$), factor banao,
phir har factor = 0 karke x nikaalo.

EXAMPLE: $x^2 + 7x + 12 = 0$
 $p+q=7$, $pq=12 \rightarrow 3$ aur 4
 $x^2 + 3x + 4x + 12 = 0 \rightarrow (x+3)(x+4)=0$
 $x = -3$ ya $x = -4$

TOPIC 3: QUADRATIC FORMULA

EXAM ALERT: Formula har quadratic pe chalta hai. Ratta maar:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Yahan $b^2 - 4ac$ ko DISCRIMINANT (D) kehte hain.

TOPIC 4: DISCRIMINANT ($D = b^2 - 4ac$)

EXAM ALERT: D se roots ka NATURE batana aata hai:

$D > 0 \rightarrow 2$ distinct real roots
 $D = 0 \rightarrow 2$ equal real roots (repeated)
 $D < 0 \rightarrow$ no real roots

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples poori tarah solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Speed word problem

(Exam favourite)

Ek train 360 km fixed speed se chalti hai. Speed 5 km/h zyada hoti toh 1 ghanta kam lagta. Original speed?

Let speed = x km/h.

$$360/x - 360/(x+5) = 1$$

Multiply by $x(x+5)$:

$$360(x+5) - 360x = x(x+5)$$

$$1800 = x^2 + 5x$$

$$x^2 + 5x - 1800 = 0$$

Split ($a=-1800$, $sum=5 \rightarrow 45, -40$):

$$(x+45)(x-40) = 0$$

$$x = -45 \text{ (reject) ya } x = 40$$

Original speed = 40 km/h.

Solved Example 2 - Number (reciprocal) problem

(Exam favourite)

Ek number aur uske reciprocal ka sum $10/3$ hai. Number?

Let number = x .

$$x + 1/x = 10/3$$

Multiply by $3x$:

$$3x^2 + 3 = 10x \rightarrow 3x^2 - 10x + 3 = 0$$

Split ($a=9$, $sum=-10 \rightarrow -9, -1$):

$$3x^2 - 9x - x + 3 = 0 \rightarrow (x-3)(3x-1)=0$$

$$x = 3 \text{ ya } x = 1/3.$$

Solved Example 3 - Consecutive integers

Do consecutive odd positive integers ka product 143. Numbers?

Let numbers = x aur $(x+2)$.

$$x(x+2) = 143 \rightarrow x^2 + 2x - 143 = 0$$

Split ($a=-143$, $sum=2 \rightarrow 13, -11$):

$$(x+13)(x-11) = 0 \rightarrow x = 11 \text{ (positive)}$$

Numbers = 11 aur 13.

Solved Example 4 - Quadratic formula

$$2x^2 - 7x + 3 = 0. (a=2, b=-7, c=3)$$

$$D = (-7)^2 - 4(2)(3) = 49 - 24 = 25$$

$$x = (7 \pm \sqrt{25}) / 4 = (7 \pm 5) / 4$$

$$x = 3 \text{ ya } x = 1/2.$$

Solved Example 5 - Nature of roots (D)

(Exam favourite)

$$(a) x^2 + 4x + 4 = 0: D = 16 - 16 = 0 \rightarrow \text{equal roots}$$

$$(b) 2x^2 - 3x + 5 = 0: D = 9 - 40 = -31 \rightarrow \text{no real roots}$$

$$(c) x^2 - 5x + 4 = 0: D = 25 - 16 = 9 \rightarrow 2 \text{ distinct roots}$$

Solved Example 6 - Find k (equal roots)

(Exam favourite)

Kis k ke liye $x^2 - 4x + k = 0$ ke EQUAL roots honge?

Equal roots $\rightarrow D = 0$

$$(-4)^2 - 4(1)(k) = 0 \rightarrow 16 - 4k = 0 \rightarrow k = 4.$$

Solved Example 7 - Simple factorisation

$$x^2 - 3x - 10 = 0$$

Split ($ac=-10$, $sum=-3 \rightarrow -5, 2$):

$$(x-5)(x+2) = 0 \rightarrow x = 5 \text{ ya } x = -2.$$

Solved Example 8 (EASIEST) - Pure quadratic

$$x^2 - 9 = 0$$

$$x^2 = 9 \rightarrow x = \pm 3 \text{ (yani 3 ya -3)}.$$

Q AND A TIME

Q1. CORE: kya $3x^2 - 2x + 1 = 0$ quadratic hai? a,b,c bata.

Q2. Factorisation se solve: $x^2 - 7x + 10 = 0$.

Q3. Quadratic formula se solve: $2x^2 + x - 6 = 0$.

Q4. Discriminant nikaal aur nature bata:

(a) $x^2 - 6x + 9 = 0$ (b) $x^2 + x + 1 = 0$

Q5. Word problem: do consecutive integers ka product 56 hai. Numbers nikaal.

Q6. Kis k ke liye $x^2 + kx + 9 = 0$ ke equal roots? ($D=0$)

Q7. CORE CHECK: agar $(x-2)(x+5)=0$, toh x ki values?

SUMMARY

1. Quadratic: $ax^2 + bx + c = 0$ ($a \neq 0$), max 2 roots.

2. Factorisation: middle term todo, $(..)(..)=0$, har factor=0 karke x nikaalo.

3. Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

4. Discriminant $D = b^2 - 4ac$:

$D > 0$ distinct, $D = 0$ equal, $D < 0$ no real roots.

5. Word problems: variable lo $\rightarrow$ equation banao $\rightarrow$ solve $\rightarrow$ impossible answer reject karo.

CORE (RED page) revise: coefficient a,b,c, factorisation, zero product rule, sqrt, +/- ka matlab.